



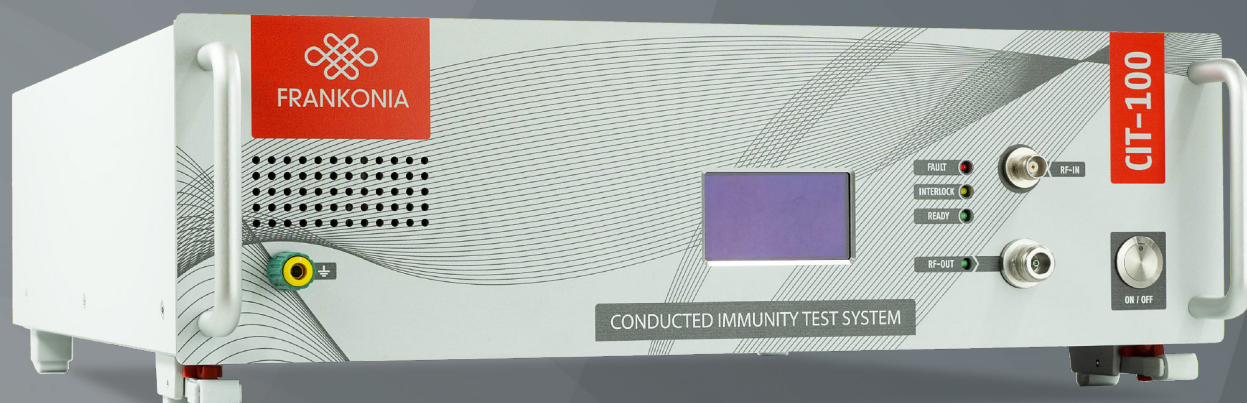
FRANKONIA

EMC Test-Systems GmbH

CIT-100

DATA SHEET

Also available as
CIT-1000*



CONDUCTED IMMUNITY TEST SYSTEM

FOR IEC 61000-4-6, ISO 11452-4 AND MIL-STD-461 CS114

*with built-in touch-screen and control PC for independent stand-alone use.

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DESCRIPTION

The CIT-100 is a complete test system for conducted RF-immunity testing and BCI-testing acc. to IEC/EN 61000-4-6, ISO 11452-4, MIL-STD 461, CS114 and similar standards.

THE SYSTEM CONSISTS OF A BUILT IN:

- ✓ Signal generator, 4kHz – 1.2 GHz
- ✓ RF-Power-Amplifier, max 4kHz – 400 MHz, 25 / 75 / 200 W
- ✓ 3-Channel RF-Power-Meter to measure the test level as well as forward & reverse power, 4kHz – 1.2 GHz
- ✓ Directional Coupler
- ✓ Comfortable control software

As a „stand-alone“ test system the CIT-100 is convincing by its easy and comfortable handling and the excellent cost-performance ratio.

We also offer the full range of coupling/decoupling networks (CDN's), EM-coupling clamp, BCI- and current clamps.



SPECIAL FEATURES

- ✓ Conducted RF immunity tests acc. to IEC/EN61000-4-6 and BCI tests acc. to ISO 11452-4 and MIL-STD 461, CS 114
- ✓ Signalgenerator, RF-power amplifier, RF-power meter and directional coupler in one 19"-case
- ✓ All built in instruments can also be used separately, via existing input / output connector.
Hence, the Signal-generator and the RF-power-meter can also be used for radiated immunity tests acc. IEC/EN 61000-4-3. Furthermore an additional external RF-Power-amplifier could be connected to the CIT-100 for this purpose.
- ✓ Stand-alone operation possible with optional available netbook
- ✓ Control-software included
- ✓ Most important parameters are shown on an integrated display
- ✓ Automatic EUT-monitoring
- ✓ Operation via USB port of a PC or Notebook
- ✓ Complete range of CDNs and EM-clamps available



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TECHN. SPECIFICATIONS	CIT-100 / 25	CIT-100 / 75 MIL	CIT-100 / 75	CIT-100 / 200
RF-Power Amplifier (TYPE)	25 W	75 W MIL	75 W	200 W
Frequency range	100 kHz-250 MHz	(4) 10 kHz-250 (400) MHz	100 kHz-400 MHz	100 kHz-400 MHz
Output Power Nominal	25 W	75 W 10W from 4 kHz - 10 kHz min. 20 W from 250 MHz - 400 MHz	75 W	200 W
Linear	20 W	50 W	50 W	100 W
Gain	46dB nominal	51dB nominal	51dB nominal	51dB nominal
Flatness	± 1.5 dB maximum			
Input power for rated input	1 mW / 0 dBm			
Input / output impedance	50 Ω			
Input VSWR	1.5:1 max			
Harmonic distortion	$<- 20$ dBc @ 20 W	$<- 20$ dBc @ 50 W	$<- 20$ dBc @ 50 W	$<- 20$ dBc @ 100 W
Noise figure	typ. 5 dB	typ. 7 dB	typ. 7 dB	tdb
Spurious output	$<- 75$ dBc @ 10 W			

TECHN. SPECIFICATIONS	CIT-100 W	CIT-100 A
Frequency range	10 kHz-400 MHz	10 kHz-400 MHz
Directional Coupler	without DC	
Frequency Range	10 kHz-1000 MHz	
Coupling	40 dB	
Max. Power	200 W	

Internal RF-Power Amplifier

Several amplifier modules are available. Highest output power can be 75 W over the specified frequency range. The amplifier input can be accessed via the back panel of the CIT-100, so that the amplifier can also be used with any external generator. 25 W and 75 W amplifiers are available as standard.

Internal RF-Voltmeter

Accurate measurements of RF signals from -40 dBm up to $+30$ dBm are done by the internal 3-channel RF-voltmeter which can be accessed (for separate use) via a BNC connector. One channel is used to measure the test level and two channels to measure the forward and reverse power via the built-in directional coupler.

Internal RF-Signal Generator

As the internal generator generates its output signal without any internal mixing components, low harmonics and spurious frequencies are assured.

Amplitude Modulation

Frequencies generated by the generator can also be modulated with a LF signal. Modulation frequencies may vary from 1 Hz up to 100 kHz, modulation levels are available from 0 % to 100 %.

User defined signals

External signals (e.g. EUT-fail or external instruments) can be connected and monitored using the application software.



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TECHNICAL SPECIFICATIONS

RF-Generator

Two switchable outputs
(Only one can be used simultaneously) 2 x SMA

Frequency range 4 kHz to 1.2 GHz

Frequency resolution 1 Hz

Output level range 0 to -63 dBm

Output level resolution 0.1 dB

Harmonics < 30 dBc

Spurious < 45 dBc

Amplitude mod. (internal) 0 - 100%, resolution 1%

Amplitude mod. (external) 0-100%, max.
Amplitude 1V=100%, BNC jack

Pulse modulation (internal) 5-95%, resolution 1%

Pulse modulation (external) DC...1MHz, 3,3/5V CMOS/TTL, BNC jack

LF-Generator (modulation)

Connector BNC jack

Frequency range 1 Hz to 100 kHz

Frequency resolution 0.1 Hz

Signal Sine wave / square wave / triangular

Amplitude 0...1 V

General Data

I/O Interface USB, LAN

Operating temperature 15° C - 40° C

Power supply

Dimensions

Weight

TECHNICAL SPECIFICATIONS

RF-Voltmeter 1 (test-level)

Connector BNC jack

Frequency range 4 kHz to 1.2 GHz

Measuring range -40 to +30 dBm

RF-Voltmeter 2 + 3 (forward and reverse power)

Connector 2 x SMA

Frequency range 4 kHz to 1.2 GHz

Measuring range -40 to +33 dBm
+ directional coupler typ. 40 dB

EUT-Monitor input

Input voltage 0-10 V

Resolution 2.5 mV

Input impedance 100 k

EUT-failed input

Input signal 3.3/5V CMOS/TTL level

Detection Mode Status or edge controlled

Temperature measurement) 10 to 100 °C (1039-1385Ω)
resolution < 1 °C (PT1000)

SCPI interfaces

USB 2.0 USB-B

LAN, 100 Mbit RJ45

GPB (optional) Centronics

Digital I/Os

Out 4 Bit Digital out, 5V CMOS/TTL

In 4 Bit Digital in, 5V CMOS/TTL

Interlock

Closes at R < 1 k Ω



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